

Metodo de Regla Falsa

Input f(x) $x^4 - 2x^2 - 5$
 xa 0
 xb 1

iteraccion	Xa	Xb	f(Xa)	f(Xb)	Xr	f(Xr)
1	0	1	-5	-6	-5	570
2	0	-5	1	5.28366219	1.16722556	-0.77452082
3	0	1.16722556	1	-0.77452082	0.65776944	0.13358842
4	0.65776944	1.16722556	0.13358842	-0.77452082	0.73271355	0.01064853
5	0.73271355	1.16722556	0.01064853	-0.77452082	0.73860644	0.00080107
6	0.73860644	1.16722556	0.00080107	-0.77452082	0.73904929	5.9989E-05
7	0.73904929	1.16722556	5.9989E-05	-0.77452082	0.73908245	4.4909E-06
8	0.73908245	1.16722556	4.4909E-06	-0.77452082	0.73908493	3.3618E-07
9	0.73908493	1.16722556	3.3618E-07	-0.77452082	0.73908512	2.5166E-08
10	0.73908512	1.16722556	2.5166E-08	-0.77452082	0.73908513	1.8839E-09
11	0.73908513	1.16722556	1.8839E-09	-0.77452082	0.73908513	1.4103E-10
12	0.73908513	1.16722556	1.4103E-10	-0.77452082	0.73908513	1.0557E-11
13	0.73908513	1.16722556	1.0557E-11	-0.77452082	0.73908513	7.9037E-13
14	0.73908513	1.16722556	7.9037E-13	-0.77452082	0.73908513	5.9175E-14
15	0.73908513	1.16722556	5.9175E-14	-0.77452082	0.73908513	4.4409E-15
16	0.73908513	1.16722556	4.4409E-15	-0.77452082	0.73908513	0

$$x_r = x_b - \frac{f(x_b)(x_b - x_a)}{f(x_b) - f(x_a)}$$

f(Xa)f(Xr)	Tolerancia ABS
-2850	
-0.77452082	528.3662
0.13358842	77.4521
0.00142252	10.2283
8.5302E-06	0.7978
4.8056E-08	0.0599
2.6941E-10	0.0045
1.5098E-12	0.0003
8.4605E-15	0.0000
4.7411E-17	0.0000
2.6569E-19	0.0000
1.4889E-21	0.0000
8.3443E-24	0.0000
4.677E-26	0.0000
2.6279E-28	0.0000
0	0.0000